Aditya Kurup

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ABOUT

Computer Science (AI) undergraduate building practical automation with Python, LLMs, AWS, and Slack applications. Experienced in schema validated model workflows, evaluation, secure approval gates, and reproducible testing. I enjoy learning unfamiliar systems quickly, communicating technical tradeoffs clearly, and owning reliable deliverables from design through verification.

EDUCATION

Manipal Institute of Technology, Bengaluru, India | B.Tech in Computer Science (AI), Minor in Quantum Computing | 2023-2027

CGPA: 8.47/10 | **Coursework:** Machine Learning, Operating Systems, Computer Networks, Data Structures and Algorithms, DBMS, Probability, Statistics, Linear Algebra

EXPERIENCE

AI Intern, Zino Technologies - Bengaluru, Karnataka | Jun 2025 - Jul 2025

• Built Python LLM workflows that converted free-form product descriptions into schema validated JSON for a low-code B2B SaaS platform, rejecting malformed generations before downstream code generation.

• Wrote an evaluation harness comparing prompt variants and foundation models on output validity and schema conformance; used data driven results to select the shipped workflow.

Research Intern, Qomputing.ai - Bengaluru, Karnataka | Mar 2026 - Apr 2026

• Developed Python simulation backends for statevector execution, implementing low level compute paths that step circuits through gate sequences.

• Profiled batched numerical workloads, located bottlenecks in per gate update loops, and restructured hot paths to improve throughput.

PROJECTS

OpsPilot Lite - Approval Gated AWS Provisioning | GitHub | Python, Slack Bolt, DeepSeek, boto3, AWS CDK

• Built a working Slack workflow that converts natural language requests into strict JSON plans, validates them with Pydantic, and requires approval from a separate allowlisted user.

• Implemented an allowlisted boto3 S3 executor with dry run mode, encryption, public access blocking, TTL and ownership tags, audit logging, duplicate prevention, and expired empty bucket cleanup.

• Added CDK deployment infrastructure for Lambda, API Gateway, SQS, DynamoDB, EventBridge, and Secrets Manager; verified the local Slack to LLM to approval flow end to end.

Policy Gated Growth Agent | GitHub | Python, Gemini API, Razorpay API

• Designed deterministic policy and atomic reservation layers so model output could propose offers but never make money affecting decisions; validated 629 oracle cases and stress tested 5,000 concurrent attempts with zero overdraws.

LLM Self Correction Reliability Study | GitHub | Python, Pandas, NumPy, Scikit learn, SQL

• Built a reproducible evaluation pipeline covering 5,760 test events across four models and trained a question grouped logistic gate that improved pooled accuracy by 2.59 percentage points over baseline.

TECHNICAL SKILLS

Languages: Python, Go, C, Java, SQL, JavaScript | **AI/ML:** LLM APIs, prompt evaluation, schema validation, RAG, Scikit learn, TensorFlow, PyTorch, NumPy, Pandas | **Cloud & Automation:** AWS (S3, Lambda, DynamoDB, EventBridge), AWS CDK, Slack Apps, REST APIs, Git, Linux | **Exposure:** Jenkins, ServiceNow workflows, infrastructure as code, cloud security, Agile/Scrum